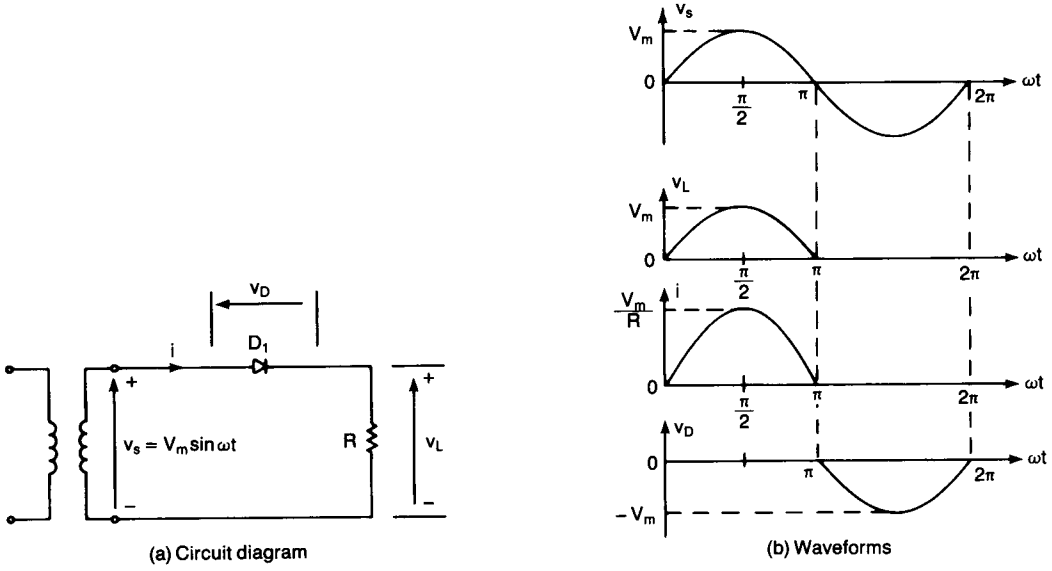


Figure 2-12 Single-phase half-wave rectifier.

2-6 SINGLE-PHASE HALF-WAVE RECTIFIERS

A *rectifier* is a circuit that converts an ac signal into a unidirectional signal. Diodes are used extensively in rectifiers. A single-phase half-wave rectifier is the simplest type and is not normally used in industrial applications. However, it is useful in understanding the principle of rectifier operation. The circuit diagram with a resistive load is shown in Fig. 2-12a. During the positive half-cycle of the input voltage, diode D_1 conducts and the input voltage appears across the load. During the negative half-cycle of the input voltage, the diode is in a *blocking condition* and the output voltage is zero. The waveforms for the input and output voltages are shown in Fig. 2-12b.

2-7 PERFORMANCE PARAMETERS

Although the output voltage as shown in Fig. 2-12b is dc, it is discontinuous and contains harmonics. There are different types of rectifier circuits and the performances of a rectifier are normally evaluated in terms of the following parameters:

The *average* value of the output (load) voltage, V_{dc}
The average value of the output (load) current, I_{dc}
The output *dc power*,

$$P_{dc} = V_{dc}I_{dc} \tag{2-42}$$

The *rms* value of the output voltage, V_{rms}
The *rms* value of the output current, I_{rms}

The output *ac power*,

$$P_{ac} = V_{rms} I_{rms} \tag{2-43}$$

The *efficiency* (or *rectification ratio*) of a rectifier, which is a figure of merit and permits us to compare the effectiveness, is defined as

$$\eta = \frac{P_{dc}}{P_{ac}} \tag{2-44}$$

The output voltage can be considered as being composed of two components: (1) the dc value, and (2) the ac component or ripple.

The *effective* (rms) value of the ac component of output voltage is

$$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2} \tag{2-45}$$

The *form factor*, which is a measure of the shape of output voltage, is

$$FF = \frac{V_{rms}}{V_{dc}} \tag{2-46}$$

The *ripple factor*, which is a measure of the ripple content, is defined as

$$RF = \frac{V_{ac}}{V_{dc}} \tag{2-47}$$

Substituting Eq. (2-45) in Eq. (2-47), the ripple factor can be expressed as

$$RF = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} = \sqrt{FF^2 - 1} \tag{2-48}$$

The *transformer utilization factor* is defined as

$$TUF = \frac{P_{dc}}{V_s I_s} \tag{2-49}$$

where V_s and I_s are the rms voltage and rms current of the transformer secondary, respectively.

If ϕ is the angle between the fundamental components of the input current and voltage, ϕ is called the *displacement angle*. The *displacement factor* is defined as

$$DF = \cos \phi \tag{2-50}$$

The *harmonic factor* of the input current is defined as

$$HF = \left(\frac{I_s^2 - I_1^2}{I_1^2}\right)^{1/2} = \left[\left(\frac{I_s}{I_1}\right)^2 - 1\right]^{1/2} \tag{2-51}$$

where I_1 is the fundamental rms component of the input current. The input *power factor* is defined as

$$PF = \frac{V_s I_1}{V_s I_s} \cos \phi = \frac{I_1}{I_s} \cos \phi \tag{2-52}$$

Note. If the input current is purely sinusoidal, $I_1 = I_s$ and the power factor, PF, equals the displacement factor, DF. An ideal rectifier should have: $\eta = 100\%$, $V_{ac} = 0$, FF = 0, RF = 0, TUF = 1, HF = 0, and PF = 1.

Example 2-6

The rectifier in Fig. 2-12a has a purely resistive load of R. Determine the (a) efficiency; (b) form factor; (c) ripple factor; (d) transformer utilization factor; and (e) peak inverse voltage (PIV) of diode D_1.

Solution The average output voltage V_{dc} is defined as

$$V_{dc} = \frac{1}{T}\int_0^T v_L(t)\,dt \tag{2-53}$$

We can notice from Fig. 2-12b that $v_L(t) = 0$ for $T/2 \le t \le T$. Hence we have

$$V_{dc} = \frac{1}{T}\int_0^{T/2} V_m \sin \omega t\,dt = \frac{-V_m}{T}\left(\cos\frac{\omega T}{2} - 1\right)$$

But the frequency of the source is $f = 1/T$ and $\omega = 2\pi f$.

$$\begin{aligned} V_{dc} &= \frac{V_m}{\pi} = 0.318V_m \\ I_{dc} &= \frac{V_{dc}}{R} = \frac{0.318V_m}{R} \end{aligned} \tag{2-54}$$

The *root-mean-square* (rms) value of a periodic waveform is defined as

$$V_{rms} = \left[\frac{1}{T}\int_0^T v_L^2(t)\,dt\right]^{1/2} \tag{2-55}$$

For a sinusoidal voltage of $v_L(t) = V_m \sin \omega t$ for $0 \le t \le T/2$, the rms value of the output voltage is

$$\begin{aligned} V_{rms} &= \left[\frac{1}{T}\int_0^{T/2} (V_m \sin \omega t)^2\,dt\right]^{1/2} = \frac{V_m}{2} = 0.5V_m \\ I_{rms} &= \frac{V_{rms}}{R} = \frac{0.5V_m}{R} \end{aligned} \tag{2-56}$$

From Eq. (2-42), $P_{dc} = (0.318V_m)^2/R$, and from Eq. (2-43), $P_{ac} = (0.5V_m)^2/R$.

(a) From Eq. (2-44), the efficiency, $\eta = (0.318V_m)^2/(0.5V_m)^2 = 40.5\%$.

(b) From Eq. (2-46), the form factor, FF $= 0.5V_m/0.318V_m = 1.57$ or 157%.

(c) From Eq. (2-48), the ripple factor, RF $= \sqrt{1.57^2 - 1} = 1.21$ or 121%.

(d) The rms voltage of the transformer secondary is

$$V_s = \left[\frac{1}{T}\int_0^T (V_m \sin \omega t)^2\,dt\right]^{1/2} = \frac{V_m}{\sqrt{2}} = 0.707V_m \tag{2-57}$$

The rms value of the transformer secondary current is the same as that of the load:

$$I_s = \frac{0.5V_m}{R}$$

The *volt-ampere* rating (VA) of transformer, VA = $V_s I_s = 0.707 V_m \times 0.5 V_m/R$.
From Eq. (2-49) TUF = $0.318^2/(0.707 \times 0.5) = 0.286$.

(e) The peak reverse (or inverse) blocking voltage, PIV = V_m.

Note. 1/TUF = 1/0.286 = 3.496 signifies that the transformer must be 3.496 times larger than that when it is being used to deliver power from a pure ac voltage. This rectifier has a high ripple factor, 121%; a low efficiency, 40.5%; and a poor TUF, 0.286. In addition, the transformer has to carry a dc current, and this results in a dc saturation problem of the transformer core.